```
for num in range(1,51):
    if num%3 == 0 and num%5 == 0:
        print("FizzBuzz")
    elif num%3 == 0:
        print("Fizz")
    elif num%5 == 0:
        print("Buzz")
    else:
        print(num)
```

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
Fizz
28
29
FizzBuzz
31
32
Fizz
34
Buzz
Fizz
37
38
Fizz
Buzz
41
Fizz
43
44
FizzBuzz
46
47
Fizz
49
Buzz
```

```
def leap_year(year):
    if year%400==0:
        return True
    elif year%100 == 0:
        return False
    elif year%4 == 0:
        return True
    else:
        return False

year = int(input("Enter year: "))
print(leap_year(year))
```

```
Enter year: 2012
True
```

```
L1 = [1,2,3]
L2 = [1,2,3]
print(L1==L2)
print(L1 is L2)
```

```
True
False
```

```
a = 5
b = 9
a = a^b
b = a^b
a = a^b
print("a=",a)
print("b=",b)
```

```
a= 9
b= 5
```

```
num = 2
count = 0
while count < 10:
    prime=True
    for i in range(2,num):
        if num%i == 0:
            prime = False
            break
    if prime:
        print(num)
        count = count+1
    num=num+1
```

```
2
3
5
7
11
13
17
19
23
29
```

```
for i in range(1,21):
    if i%4 == 0:
        pass
    if i==13:
        continue
    if i==18:
        break
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
11
12
14
15
16
17
```

```
marks = int(input("Enter marks: "))
if marks >= 90:
    print("A")
elif marks >= 80:
    print("B")
elif marks >= 70:
    print("C")
else:
    print("Fail")
```

```
Enter marks: 97
A
```

```
num = int(input("Enter number: "))
reverse = 0
while num > 0:
    digit = num%10
    reverse = reverse*10+digit
    num = num//10
print(reverse)
```

```
Enter number: 5678
8765
```

```
scores = [50, 80, 90, 70, 60]
highest = scores[0]
second = scores[0]
for score in scores:
    if score > highest:
        second = highest
        highest = score
    elif score > second and score != highest:
        second = score
print("Second highest:", second)
```

```
Second highest: 80
```

```
numbers = [1,2,2,3,3,4]
unique = list(set(numbers))
print(unique)
```

```
[1, 2, 3, 4]
```

```
list1 = [1,2,3,4]
list2 = [3,4,5,6]
common = []
for i in list1:
    if i in list2:
        common.append(i)
print(common)
```

```
[3, 4]
```

```
t = (10,20,30)
try:
    t[1] = 50
except:
    print("Tuple cannot be changed")
```

```
Tuple cannot be changed
```

```
text = input("Enter a string: ")
count = {}
for ch in text:
    if ch in count:
        count[ch] = count[ch] + 1
    else:
        count[ch] = 1
print(count)
```

```
Enter a string: varun
{'v': 1, 'a': 1, 'r': 1, 'u': 1, 'n': 1}
```

```
squares = [i * i for i in range(1, 21) if i % 2 == 0]
print(squares)
```

```
[4, 16, 36, 64, 100, 144, 196, 256, 324, 400]
```

```
data = {'a': 1, 'b': 2, 'c': 3, 'd': 4}
result = {key: value for key, value in data.items() if value > 2}
print(result)
```

```
{'c': 3, 'd': 4}
```